Multi-Feature Weighted Sparse Graph for SAR Image Analysis

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***Abstract*— Sparse representation (SR) method has the advantages of good category distinguishing performance, noise robustness, and data adaptiveness. In this article, a multi-feature weighted sparse graph (MWSG) is presented for synthetic aperture radar (SAR) image analysis. First, multiple types of features are extracted to fully describe the characteristics of SAR image. Then, multiple SRs of samples in multiple feature spaces are obtained by solving a weighted joint SR model, in which the weight is the Gaussian kernel distance among samples. Moreover, a new fusion mechanism is given to integrate multiple weighted SRs, which aims to eliminate the negative influence of the singular data, so the MWSG is obtained. Afterward, the brief steps of the SAR image segmentation and semisupervised classification based on MWSG are stated. A series of experiments on the simulated and real SAR images shows that the MWSG has better performance than other existing relevant methods.**

***Index Terms*— Semisupervised classification, synthetic aperture radar (SAR) image segmentation, weighted sparse representation (SR).**

I. Introduction

SYNTHETIC aperture radar (SAR) is an active imaging system with microwave, which can work day and night under all weather conditions [1], and peer through trees, clouds, and other obstacles to obtain an image with high resolution. It has been widely used in various fields, such as military target recognition, resource exploration, forest survey, topographic mapping, and disaster monitoring. Until now, many researchers have done a great deal of studies on SAR image analysis and understanding. As the essential steps of SAR image analysis and understanding, SAR image segmentation [2] and classification [3] are still challenging tasks. Image segmentation aims to divide an image into some meaningful and nonoverlapping regions, and image classification is to classify per image pixel or region into one of the groups with different characteristics.

So far, many methods have been applied in SAR image segmentation and classification, such as thresholding [4], fuzzy clustering [5], Markov random field [6], support vector machine [7], neural network [8], immune clone [9], and sparse representation (SR) [10]. Comparing with conventional approaches, the emerging SR technology has mainly three advantages. Our previous works [11] have indicated that the SR method using data set itself as the dictionary is actually equivalent with weighted nonlocal denoising algorithm. It shows that the SR algorithm has the robustness to data noise, which can decrease the noise of the SAR image without the filter preprocessing. By contrast, the conventional approaches usually utilize filtering algorithm to reduce the inherent speckle noise of SAR image before segmenting and classifying the SAR image. It will cause that some effective information is inevitably lost and, thus, results in blurry edges and textures, which is not conducive to obtain the final accurate SAR image segmentation and classification results. SAR image segmentation and classification methods based on SR can gradually decrease the noise of SAR image and well preserve edges and textures of SAR images in the iterative process. Moreover, the dictionary of the SR model can consist of sample set itself [12]. Thus, it can be avoided that a large amount of SAR image samples are gathered by consuming vast manpower and time costs. Meanwhile, this characteristic of the SR method makes up the defect about lack of SAR image samples. Furthermore, when using the sample set itself as a dictionary in SR, the SR method has the advantage of data adaptiveness. It avoids that the difference between the pending SAR images with the gathered SAR image samples leads to the reduction of the final SAR image segmentation and classification accuracy rates. Therefore, in this article, SAR image segmentation and classification algorithms based on SR are concerned. In addition, since the labeled samples in

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the SAR image are unavailable, a semisupervised classification method [13] which only needs to use a small number of labeled samples is adopted in this article.

At first, a sparse representation-based classification (SRC) method presented by Wright *et al.* [14] attracted much attention because of its favorable classification result, which is better than other conventional classification algorithms. During the same period, Elhamifar and Vidal [15] presented a sparse subspace clustering (SSC) algorithm, which successfully applied the SR method to separate different moving objects in video. And then, an L1-graph [16] was proposed, which is similar to the model in SSC algorithm. Afterward, a method of low-rank representation (LRR) [17] is presented to segment the image. On the basis of LRR and L1-graph, a low-rank graph (LR_G) [18] was proposed for semisupervised classification. In the sequel, a weighted sparse representation for classification (WSRC) [19] was proposed, which exploited the distance information in representing the test samples to improve the classification effectiveness of SRC. The representation result of WSRC is sparser than that of SRC and can obtain better recognition results. It inspires us to exploit the weighted way to enhance the performance of the SR model. Furthermore, the existing weighted sparse representation methods are applied to face recognition [19] and hyperspectral image classification [20], which belongs to the supervised classification problem. In this article, the weighted sparse representation method is used for image segmentation and semisupervised classification.

In addition, image feature obtained by transformation, statistics, or any other is used to describe the essential attribute of the image, so feature extraction plays a necessary role in image analysis. Since different features could potentially belong to different feature spaces, a large number of researchers utilized many kinds of features [21], [22] to fully and accurately depict image characteristics. Multiple features are fused by concatenation or the sum of the squares in most of the existing methods [23], [24]. However, the contributions of different types of features to image characteristic are different. Gu *et al.* [11] have applied the SR method and multi-feature into SAR image segmentation and semisupervised classification tasks in the previous works. However, it did not take difference among multiple SR coefficients corresponding to multi-feature into consideration.

In this article, we propose a new algorithm called multi-feature weighted sparse graph (MWSG) and apply it to segment and classify SAR image. First, we divide a SAR image into a large number of homogeneous regions and extract several kinds of features of every region to get several feature sets. Then, we compute the distances among the samples and exploit the distance information to define the weights of the SR coefficients. The SR coefficient matrices corresponding to different feature sets are obtained by solving a new weighted joint SR model. Moreover, a novel way is adopted to fuse the SR coefficients in multi-feature sets and, thus, form the adjacent matrix of MWSG. Finally, we use spectral clustering [25] to partition the vertices of graph and obtain the SAR image segmentation result, and semisupervised classification method-based graph to get the SAR image classification result.

The remaining parts of this article are organized as follows. Section II reviews the initial SR model and existing SR-based image segmentation and classification methods. A new weighted joint SR model and the MWSG are introduced in Section III. Section IV presents the MWSG-based SAR image segmentation and semisupervised classification algorithms. Experimental results in different SAR images that assess the performance of the algorithms are given in Section V. Finally, Section VI concludes this article.

II. Related Work

A. Sparse Representation

The simplest SR model is as follows:

$$\min_{\mathbf{z}} \|\mathbf{z}\|_0 \quad \text{s.t. } \mathbf{x} = D\mathbf{z} \tag{1}$$

where $\mathbf{x}$ is an m-dimensional sample, D is an $m \times d$ ($m \ll d$) dictionary, $\mathbf{z} \in R^d$ is the SR coefficient of $\mathbf{x}$ in D, and $\|\cdot\|_0$ is ℓ_0-norm of a vector, the meaning of which is the number of nonzero elements in the vector. The model in (1) indicates that a sample can be represented by a linear combination of small amounts of atoms in a dictionary.

Generally, since the objective function in (1) is an NP-hard problem [26]–[28], it is often replaced with

$$\min_{\mathbf{z}} \|\mathbf{z}\|_1 \quad \text{s.t. } \mathbf{x} = D\mathbf{z} \tag{2}$$

where $\|\cdot\|_1$ is ℓ_1-norm of a vector, and $\|\mathbf{z}\|_1 = \sum_i |z_i|$.

For a sample set X that contains n samples $\mathbf{x}_1, \ldots, \mathbf{x}_n$, the SR of X can be obtained by

$$\min_{Z} \|Z\|_1 \quad \text{s.t. } X = DZ \tag{3}$$

where Z is the SR matrix of a sample set X, the ith column $\mathbf{z}_i$ of which is the SR coefficient of the ith sample $\mathbf{x}_i$ in X.

B. SR-Based Image Classification and Segmentation Methods

At first, the SR theory was applied to facial recognition, and an SRC method was proposed by Wright *et al.* [14]. The SRC algorithm is to use a sparse linear combination of face images from the training set to represent a test face image. Since then, the image classification method based on SR is further developed from two primary aspects. The one is that some new SR models were proposed for face recognition [29], [30]. The other is that the SRC algorithm was applied in other types of images, such as hyperspectral image [31] and ear image [32]. Overall, image classification approaches based on SR mainly include the following steps: 1) construct dictionary by using a large amount of training samples from different classes; 2) solve an SR model to get an SR coefficient of a test sample in the dictionary; 3) compute the residuals between the test sample and its estimations in every category by utilizing the SR coefficient; and 4) determine the category label of the test sample according to the minimum residual.

Moreover, many researchers have been working on segmentation problem based on the SR technology. An SSC [15] algorithm was presented, which combined the SR theory with spectral graph segmentation method to separate different moving objects in video. After then, Gu *et al.* [11] used the SR

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technic to segment a single image [11]. The main procedure of the image segmentation based on SR is as follows: 1) oversegment an input image into many superpixels; 2) extract feature of every superpixel; 3) obtain the SR coefficient of each feature set by solving an SR model; and 4) use spectral clustering algorithm to get the segmentation result in the case of treating the SR coefficient as the adjacency weights of graph.

In addition, semisupervised classification method based on SR was present [33], [34]. It utilizes many unlabeled samples and a small amount of labeled samples as the dictionary of SR, which is different with the classification based on SR. And then, similarly with the image segmentation method based on SR, the obtained SR coefficient of all samples is regarded as the adjacency weights of graph. Finally, the little category information is propagated to the unlabeled samples by the weights of graph.

III. Multi-Feature Weighted Sparse Graph

A. Weighted Joint Sparse Representation

The WSRC algorithm used the distances between the training samples and a given test sample to determine the weights of the training samples. Unlike the WSRC, the distance information among samples is used to define the weight of the SR coefficient in this article. We adopt the Gaussian kernel to measure the distance among samples since the Gaussian kernel distance can capture the nonlinear information within the data set and describe accurately the similarity among samples, which is similar to the WSRC method. For any two samples $\mathbf{x}_i$ $(1 \le i \le n)$ and $\mathbf{x}_j$ $(1 \le j \le n)$ in the data set $X = \{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$, their distance is calculated as follows:

$$w(\mathbf{x}_i, \mathbf{x}_j) = \exp(-\|\mathbf{x}_i - \mathbf{x}_j\|^2 / 2\sigma^2) \tag{4}$$

where σ is the Gaussian kernel width. The more similar the two samples $\mathbf{x}_i$ and $\mathbf{x}_j$ are, the larger the corresponding weightings $w(\mathbf{x}_i, \mathbf{x}_j)$ and $w(\mathbf{x}_j, \mathbf{x}_i)$ are.

The Gaussian kernel distance between every two samples in the data set is computed and used to construct an $n \times n$ matrix $W = \left\{ w_{ij} \middle| 1 \le i \le n, 1 \le j \le n \right\}$, where w_{ij} is the short form of $w(\mathbf{x}_i, \mathbf{x}_j)$. Since the Gaussian kernel distance w_{ij} is between 0 and 1, the matrix W which contains the distance information between samples is directly used as the weight of the SR coefficient matrix. Then, we exploit the weight matrix and give a new weighted joint SR model

$$\min_{Z^1, \ldots, Z^k} \sum_{b=1}^{k} \|W^b \cdot Z^b\|_1 + \lambda \|X^b - X^b Z^b\|_{\mathrm{F}}^2 \text{ s.t. } \operatorname{diag}(Z^b) = 0 \tag{5}$$

where $X^1, \ldots, X^k$ denote the k types of feature sets of the data set X; Z^b is the SR coefficient of X^b, and $\|Z^b\|_1 = \sum_{i,j} |Z^b_{ij}|$; W^b denotes the weight of Z^b; $W^b \cdot Z^b$ denotes the point multiplication of matrices W^b and Z^b; $\|\cdot\|_{\mathrm{F}}$ is the Frobenius norm; and $\lambda > 0$ is the parameter to balance the effect of different parts. The constraint $\operatorname{diag}(Z^b) = 0$ is to avoid that the solution of (5) is the identity matrix.

It follows from the principle of SR that the number of rows in X^b as the dictionary is much smaller than the number of columns, so the objective function in (5) does not have the closed-form solution. It can be verified by simplifying the problem ($k = 1$) in (5) and supposing X^1 as a 1×3 matrix. In this article, an alternating direction method of multipliers (ADMM) [35] is used to solve the objective function in (5), since the problem in (5) is divided into some subproblems that have closed-form solutions by using the ADMM method. First, a set of auxiliary variables $H^1, \ldots, H^k$ are introduced; thus, the objective function in (5) is converted into equivalent form as follows:

$$\min_{\substack{H^1, \ldots, H^k \\ Z^1, \ldots, Z^k}} \sum_{b=1}^{k} \|W^b \cdot H^b\|_1 + \lambda \|X^b - X^b Z^b\|_{\mathrm{F}}^2 \quad \text{s.t. } Z^b = H^b, \quad \operatorname{diag}(H^b) = 0.$$

And then, minimize the following augmented Lagrange function:

$$\sum_{b=1}^{k} \left(\|W^b \cdot H^b\|_1 + \lambda \|X^b - X^b Z^b\|_{\mathrm{F}}^2 \right) + \sum_{b=1}^{k} \left(\left\langle P^b, Z^b - H^b \right\rangle + \frac{\mu}{2} \|Z^t - H^t\|_{\mathrm{F}}^2 \right)$$

where $P^1, \ldots, P^k$ are the Lagrange multipliers and $\mu > 0$ is a penalty parameter. Algorithm 1 describes the solution procedure of (5) in detail.

In Algorithm 1, the objective function in (6) can be solved by the soft thresholding operator [36].

According to the analysis of the existing related works [11], the magnitude of the SR coefficients can reveal the similarity among the samples. The larger the SR coefficient is,

Algorithm 1 Solving the Weighted Joint Spars Representation Model by ADMM

Input: multiple feature sets $X^1, \ldots, X^b, \ldots, X^k$ $(b = 1, \ldots, k)$, parameters λ.

Initialize: convergence threshold $\eta = 10^{-5}$, penalty parameter $\mu = 10^{-6}$ and enhancement factor $\theta = 1.1$, $Z^b = H^b = P^b = 0$.

Iterate: if $\|Z^b - H^b\|_\infty < \eta$

1. Update $H^1, \ldots, H^b, \ldots, H^k$:

$$\arg\min_{H^b} \frac{W^b}{\mu} \left\| H^b \right\|_1 + \frac{1}{2} \left\| H^b - \left(Z^b + \frac{P^b}{\mu} \right) \right\|_F^2. \tag{6}$$

2. Update $Z^1, \ldots, Z^b, \ldots, Z^k$:

$$Z^b = \left((X^b)^T X^b + \frac{\mu}{2\lambda} I \right)^{-1} \left((X^b)^T X^b + \frac{\mu}{2\lambda} H^b - \frac{1}{2\lambda} P^b \right),$$

where I is an $n \times n$ identity matrix and T denotes transpose of matrix.

3. Update the multipliers: $P^b = P^b + \mu \left(Z^b - H^b \right)$.

4. Update the parameters: $\mu = \min \left(\theta \mu, 10^{10} \right)$.

Output: k sparse representation matrices $Z^1, \ldots, Z^b, \ldots, Z^k$.

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the more similar the corresponding two samples are. It is the same with the weighting obtained by the Gaussian kernel function. Hence, through the constraint of the weighting matrix, the weighted SR coefficients can enlarge the similarity among the samples from the same class and difference among the samples belonging to different classes. Thus, the category distinguishing ability of SR coefficients obtained by (5) is improved, which is conducive to the subsequent clustering and classification tasks to obtain better clustering and classification results than using the unweighted SR coefficients.

B. Multi-Feature Weighted Sparse Graph

As mentioned earlier, multiple SR coefficient matrices are integrated by the sum of the squares way in the existing method. However, due to the influence of the noise or the similar characteristic of samples from different classes, the singular data are possible. To eliminate the negative effect of the singular data to the final clustering or classification result, it can be removed and the remaining data are reserved for the integration.

For the SR coefficient matrices $Z^1, \ldots, Z^b, \ldots, Z^k$ corresponding to different feature sets, we first calculate the mean value of each SR coefficient matrix. Then, the only one which is greater or less than the corresponding mean value in multiple SR coefficients with the same position in the matrix is removed. After that, the norm of vector formed by, respectively, multiplying the remaining SR coefficients by corresponding weightings is computed as the integration SR coefficient. Suppose that $m^1, \ldots, m^b, \ldots, m^k$ denote the mean values of the SR coefficient matrices $Z^1, \ldots, Z^b, \ldots, Z^k$, respectively. Thus, the weighted integration SR coefficient matrix $Z = \left\{ z_{ij} \middle| 1 \le i \le n, 1 \le j \le n \right\}$ is computed as follows:

$$z_{ij} = \begin{cases} \sqrt{\sum_{b=1}^{k} \left(w_{ij}^b z_{ij}^b\right)^2 - \left(w_{z_\min} z_{\min}\right)^2}, \\ \qquad \text{if } \sum_{b=1}^{k} \delta\left(z_{ij}^b - m^b\right) = k-1 \\ \sqrt{\sum_{b=1}^{k} \left(w_{ij}^b z_{ij}^b\right)^2 - \left(w_{z_\max} z_{\max}\right)^2}, \\ \qquad \text{if } \sum_{b=1}^{k} \delta\left(z_{ij}^b - m^b\right) = 1 \\ \sqrt{\sum_{b=1}^{k} \left(w_{ij}^b z_{ij}^b\right)^2}, \quad \text{otherwise.} \end{cases} \tag{7}$$

where $z_{\min}$ is the only one that is less than the corresponding mean value in $z_{ij}^1, \ldots, z_{ij}^b, \ldots, z_{ij}^k$, and $w_{z_\min}$ is the weight corresponding to $z_{\min}$; similarly, $z_{\max}$ is the only one that is greater than the corresponding mean value in $z_{ij}^1, \ldots, z_{ij}^b, \ldots, z_{ij}^k$, and $w_{z_\max}$ is the weight corresponding to $z_{\max}$; obviously, $k \ge 3$; $\delta(\cdot)$ is a sign function which is defined as follows:

$$\delta(x) = \begin{cases} 1, & x \ge 0 \\ 0, & x < 0. \end{cases} \tag{8}$$

We think that the only one which is greater or less than the corresponding mean value in multiple SR coefficients is the singular data, since it has low rate of occurrence. The singular data are removed by using (7), and then, its negative influence on the final clustering or classification result is weakened. That is, the classification performance of the integration SR coefficient is improved by eliminating the singular data.

After eliminating the effect of the singular data, the weighted integration SR coefficient matrix Z is regarded as the adjacent matrix of graph. Thus, the MWSG is obtained. Overall, the MWSG enhances the category distinguishing the performance of the typical SR coefficient from two aspects. One is the usage of weight, and another one is the elimination of the singular data in a new fusion mechanism.

C. Computational Complexity Analysis

In this section, we analyze the computational complexities of MWSG and several common graph based on SR: L1 graph (L1_G) [33], LR_G [34], and multi-task sparse representation graph (MSR_G) [28]. Suppose that the number of iterations is t, the time complexity of L1_G is $O\left(n^2 t\right)$. The computing complexity of LR_G is $O\left(n^3 t + n^2 t\right)$ since the model of LR_G contains a sparse term and a low-rank term. The model in MSR_G can handle multiple tasks simultaneously, which is similar to MWSG, so the time complexity of MSR_G is $O\left(kn^2 t\right)$. As the proposed MWSG needs to calculate k matrices with a size of $n \times n$ as the weights before the iteration, its computational complexity is $O\left(kn^2 + kn^2 t\right)$. Since k is generally in single digits, the time complexity of MWSG has the same order as L1_G and MSR_G, and less than that of LR_G.

IV. SAR Image Analysis With MWSG

According to the analyses above, the obtained adjacent matrix of MWSG has good category distinguishing performance. In this article, we apply the MWSG in SAR image segmentation and semisupervised classification tasks.

Before implementing the MWSG-based SAR image segmentation and semisupervised classification, the SAR image needs to be preprocessed. The pretreatment contains two steps. The first step is to over-segment the SAR image into a host of homogeneous regions called superpixels. It not only is helpful to capture structural information of the image but also can greatly reduce the computational complexity of the algorithm since the number of superpixels is greatly less than the image pixels. The second step is to extract different types of features of the superpixels. In this article, we use three types of features: gray histogram, local gradient ratio pattern histogram (LGRPH) [37], and Gabor filter banks. Herein, the gray histogram and LGRPH features reflect the change of image brightness from different perspectives. The LGRPH feature derives from the ratio of average which is robust to speckle noise in the SAR images and was specifically proposed

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Algorithm 2 MWSG based SAR Image Segmentation

Input: multiple feature sets $X^1, \ldots, X^b, \ldots, X^k$ of SAR image, number of clusters c.

1. Compute k weighted matrices $W^1, \ldots, W^b, \ldots, W^k$ for k feature sets $X^1, \ldots, X^b, \ldots, X^k$ by (3);
2. k sparse representations $Z^1, \ldots, Z^b, \ldots, Z^k$ corresponding to k feature sets $X^1, \ldots, X^b, \ldots, X^k$ are obtained by solving the weighted joint sparse representation model according to Algorithm 1.
3. The k sparse representation matrices $Z^1, \ldots, Z^b, \ldots, Z^k$ are fused into the weighted integration sparse representation coefficient matrix Z by (6).
4. Compute the adjacent matrix of MWSG Z.
5. The MWSG is transformed into undirected graph by $\frac{1}{2}\left(Z + Z^T\right)$.
6. Use NCuts algorithm [39] in spectral clustering to partition n vertices of the undirected graph into c classes.

Output: category vector of n samples.

to describe a local feature of SAR image. The Gabor filters banks were often used to extract the texture feature of SAR image [38]. In addition, one kind of features of all image pixels in a superpixel is similar since every superpixel is a homogeneous region. The mean value of the feature of all image pixels from a superpixel is used as the corresponding feature of this superpixel. Thus, the multiple feature sets $X^1, \ldots, X^b, \ldots, X^k$ ($k = 3$ in this article) of the superpixels are obtained. Then, the MWSG of multiple feature sets is learned by the method described in Section III-B and applied to SAR image segmentation and semisupervised classification.

A. SAR Image Segmentation Based on MWSG

In the existing SAR image segmentation methods, spectral clustering algorithm has often been used in recent years, as it can be implemented simply and has good segmentation result. Algorithm 2 describes the steps of MWSG-based SAR image segmentation algorithm in detail.

B. SAR Image Semisupervised Classification Based on MWSG

Recently, many researchers pay their attention to SAR image semisupervised classification methods [40], [41], as they only need a small amount of labeled samples and can obtain favorable classification result. Simultaneously, it just makes up the lack of labeled samples in the SAR image. We get the small number of labeled samples of SAR image in the man–machine interactive way. The specific process of obtaining the labeled samples has been described in [11] and is not repeated here. The MWSG-based SAR image semisupervised classification method is given in Algorithm 3.

Algorithm 3 MWSG based SAR Image Semisupervised Classification

Input: multiple feature sets $X^1, \ldots, X^b, \ldots, X^k$ of SAR image, number of clusters c, a $n \times c$ (where n is the number of samples) initially labeled 0-1 matrix F ($F_{im} = 1$ ($1 \le i \le n$, $1 \le m \le c$) if the ith sample is labeled as the mth class, otherwise $F_{ia} = 0$), parameter $\alpha \in (0, 1)$.

1. Compute k weighted matrices $W^1, \ldots, W^b, \ldots, W^k$ for k feature sets $X^1, \ldots, X^b, \ldots, X^k$ by (3);
2. k sparse representations $Z^1, \ldots, Z^b, \ldots, Z^k$ corresponding to k feature sets $X^1, \ldots, X^b, \ldots, X^k$ are obtained by solving the weighted joint sparse representation model according to Algorithm 1.
3. The k sparse representation matrices $Z^1, \ldots, Z^b, \ldots, Z^k$ are fused into the integration sparse representation coefficient matrix Z by (6).
4. Calculate the adjacent matrix of MWSG Z.
5. Compute the final label matrix $\tilde{F} = (1-\alpha)\left(I - \alpha Z\right)^{-1} F$, where I is an $n \times n$ identity matrix.
6. Get the label of the category of the ith sample by $arg \max_{1 \le m \le c} \tilde{F}_{im}$.

Output: label vector of the category of n samples.

V. Experiment and Discussion

In this section, the performances of the proposed and other state-of-the-art algorithms are compared by a series of segmentation and semisupervised classification experiments on simulated and real SAR images. Since the proposed MWSG is one of the SR-based spectral graph methods, it first is compared with several existing SR-based construction methods of the graph: k-nearest neighbors-based graph (KNN_G) [42], locally linear embedding-based graph (LLE_G) [43], L1-graph (L1_G) [16], LR_G [18], and MSR_G [11]. In the SAR image segmentation experiments, the experimental result of MWSG is compared with that of other common clustering algorithms, such as k-means, mean shift [44], and spectral clustering. In addition, the performance of MWSG is compared with several latest SAR image segmentation and classification methods. Finally, the sensitivity of parameters in the proposed MWSG algorithm is analyzed.

A. Experimental Setting

The spectral graph method needs to calculate the relation between every two samples. If the single image pixel is regarded as a sample, the number of samples is large, so the spectral graph method involves high cost of computation and storage. To reduce the computational cost, we first transform the image pixels into superpixels by using Turbopixels [45] algorithm. Each superpixel is a homogeneous region, which contains some similar and close image pixels. Then, a certain feature of all superpixels needs to be processed so that they have the same number of dimensions. The gray histogram and LGRPH features of each superpixel are statistical features with 256-D vector. For the Gabor feature of superpixels, the convolution of the initial image and a bank of Gabor filters is first calculated, and the bank of Gabor filters is composed

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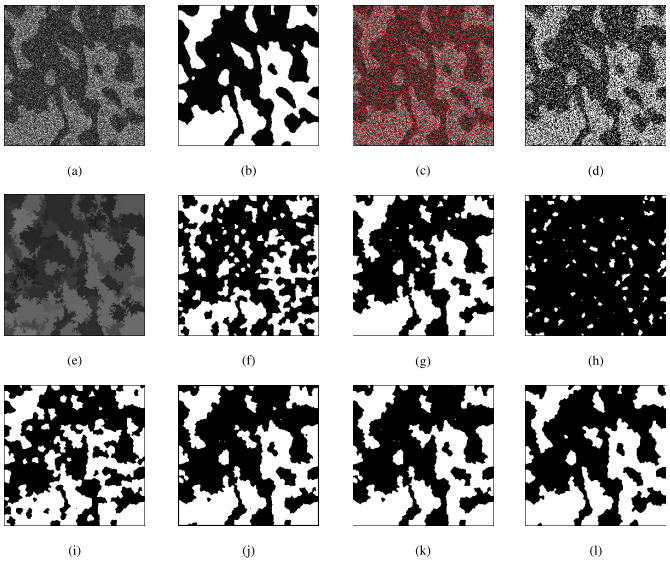


Fig. 1. Segmentation results on two-class artificial image. (a) Artificial image. (b) Ground truth. (c) Superpixels obtained by Turbopixels algorithm. (d) k-means. (e) Mean shift. (f) NCuts. (g) KNN_G. (h) LLE_G. (i) L1_G. (j) LR_G. (k) MSR_G. (l) MWSG.

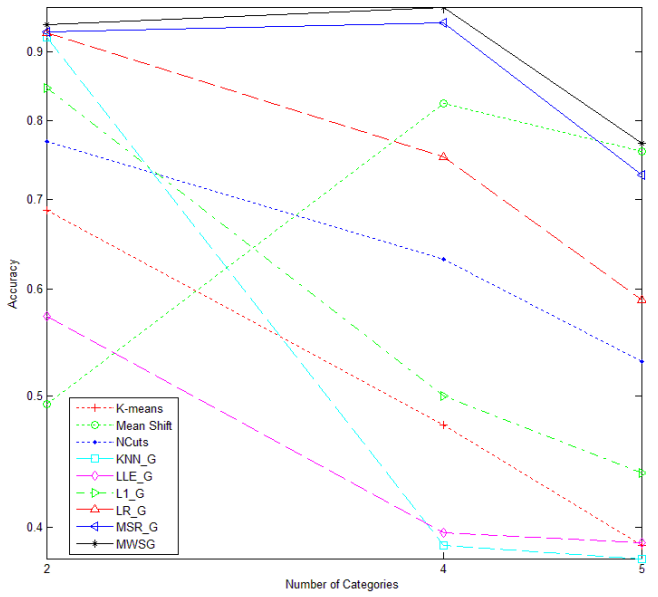


Fig. 2. Accuracy rates of the segmentations obtained by different methods for three artificial images.

of 40 filters with 5 scales and 8 orientations. Then, the mean value of the filter responds of image pixels in a superpixel is computed as a Gabor filter feature of this superpixel for each filter, so the Gabor feature of a superpixel is a 40-D vector. In addition, since the SR method can only reduce additive noise of data and the inherent speckle noise in SAR images is multiplicative, we extract features of the superpixels after the logarithmic transformation for the SAR image.

In our experiments, the number of superpixels in Turbopixels algorithm is set to 800, and the number of adjacent samples is set to 10 in the KNN_G and LLE_G. The Gaussian parameter used for the construction of graph is set to 3 in the NCuts algorithm. According to [11], the parameters α and β of MSR_G are set to 0.3 and 0.5, respectively. The parameter λ and the Gaussian kernel parameter σ of the proposed MWSG are set to 0.5 and 0.01, respectively, according to the experience. For different images, the parameters (h_s, h_r, W) of mean shift [44] are adjusted to make the visual effects and segmentation accuracy to be the best. In addition, the common NCuts [39] algorithm in the spectral clustering is used to compare with the proposed MWSG. In addition, we performed 20 runs of the k-means algorithm and retained only the best segmentation result. The other methods are executed only once since they are stable in a certain parameter.

B. Segmentation of Simulated SAR Images

To objectively evaluate the performance of the proposed MWSG in the image segmentation, three simulated three-look SAR images with the same size of 256×256 are used for experiments. The simulated SAR image has corresponding ground truth, which is used to calculate the accuracies of the segmentation results of different algorithms.

The first simulated SAR image is a two-class image, as shown in Fig. 1(a). Fig. 1(b) shows the ground truth image corresponding to Fig. 1(a), which comes from a two-class Gibbs field. The superpixels obtained by Turbopixels algorithm are shown in Fig. 1(c). Fig. 1(d) shows the segmentation result of k-means, which has many spots in consistent regions. The segmentation results of mean shift with $(h_s, h_r, W) = (15, 10, 500)$, NCuts, LLE_G, and L1_G are dreadful, as shown in Fig. 1(e), (f), (h), and (i), respectively. Although Fig. 1(g), (j), and (k) obtained, respectively, by KNN_G, LR_G, and MSR_G are better than Fig. 1(e), (f), (h), and (i), some pixels are obviously misclassified in consistent regions of Fig. 1(g), (j), and (k). In contrast, the segmentation of MWSG shown in Fig. 1(l) has good region consistency. Fig. 2 shows the segmentation accuracies of different algorithms on the simulated SAR images, which is consistent with the visual results.

Fig. 3(a) shows a four-class simulated SAR image, and the corresponding ground truth is shown in Fig. 3(b), which contains four gray levels 60, 85, 140, and 220. Similar to Fig. 1(c), Fig. 3(c) shows the superpixels of Fig. 3(a). The segmentation results of k-means, NCuts, KNN_G, LLE_G, L1_G, and LR_G are dreadful, as shown in Fig. 3(d) and (f)–(j), respectively. The four different regions in Fig. 3(e) and (k) obtained, respectively, by mean shift and MSR_G can be separated roughly. However, the segmentation of mean shift with $(h_s, h_r, W) = (15, 10, 300)$ contains 194 categories. The segmentation result of MWSG has better regional consistency than Fig. 3(k), as shown in Fig. 3(l). The segmentation accuracies of different algorithms in Fig. 3(a) are shown in Fig. 2.

Another experiment is carried out on a five-class simulated SAR image, as shown in Fig. 4(a). Fig. 4(b) has five gray levels 50, 85, 130, 180, and 220, which is the ground truth image corresponding to Fig. 4(a). The superpixels of Fig. 4(a) obtained by Turbopixels algorithm are shown in Fig. 4(c). In the segmentations of k-means, NCuts, KNN_G, LLE_G, and L1_G, a large number of pixels are misclassified, as shown in Fig. 4(d) and (f)–(i), respectively.

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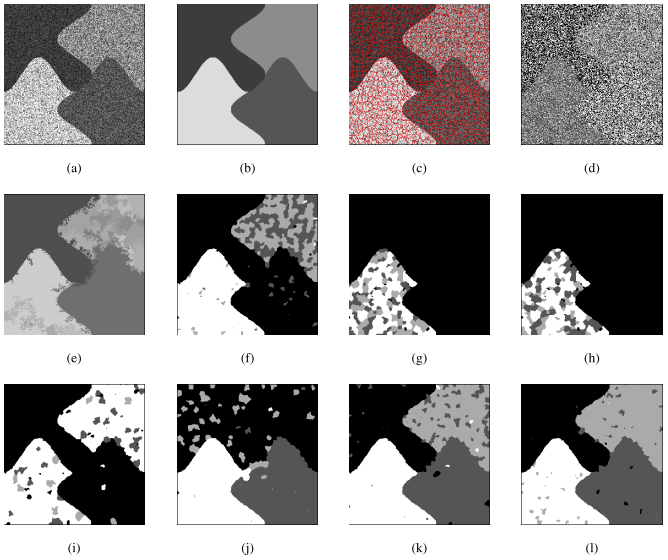


Fig. 3. Segmentation results on four-class artificial image. (a) Artificial image. (b) Ground truth. (c) Superpixels obtained by Turbopixels algorithm. (d) k-means. (e) Mean shift. (f) NCuts. (g) KNN_G. (h) LLE_G. (i) L1_G. (j) LR_G. (k) MSR_G. (l) MWSG.

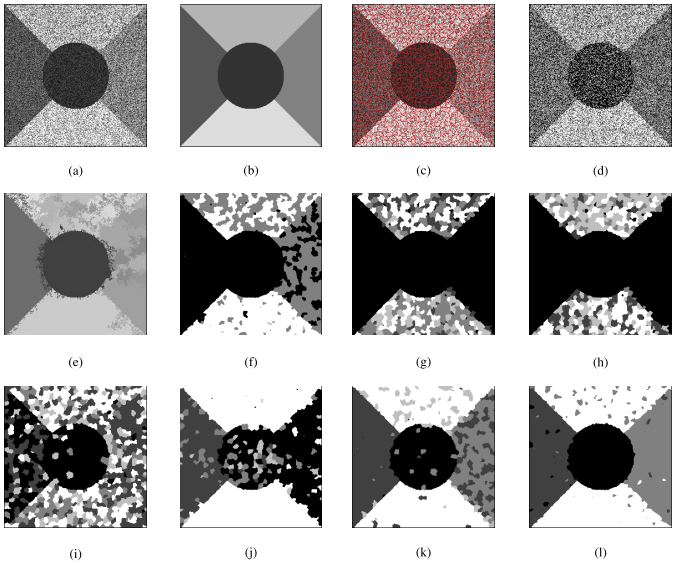


Fig. 4. Segmentation results on five-class artificial image. (a) Artificial image. (b) Ground truth. (c) Superpixels obtained by Turbopixels algorithm. (d) k-means. (e) Mean shift. (f) NCuts. (g) KNN_G. (h) LLE_G. (i) L1_G. (j) LR_G. (k) MSR_G. (l) MWSG.

Fig. 4(e), (j), and (k) shows the segmentation results of mean shift ($(h_s, h_r, W) = (15, 10, 500)$), LR_G, and MSR_G, respectively, which roughly divide Fig. 4(a) into five different regions. However, the regional consistency of Fig. 4(e), (j), and (k) is bad. The regional consistency of Fig. 4(l) is improved, which is obtained by MWSG. The corresponding segmentation accuracies are shown in Fig. 2, in which the segmentation accuracy of the proposed MWSG is highest.

In addition, Fig. 5 shows the runtime of these algorithms on the 256×256 simulated SAR image to assess their computational cost. Since the sizes of Figs. 1, 3, and, 4(a) are the same, the runtimes of the same algorithm on the three simulated SAR images are similar. The runtime of each algorithm in Fig. 5 is the mean value of runtimes of every method on three simulated SAR images. In addition, except the mean shift, other methods are performed in MATLAB R2010a platform of a personal computer with 3.20 GHz Intel Core i3 processors, 3-GB memory, and Windows XP operating system. The mean shift algorithm runs in Visual Studio, code of which is available at http://coewww.rutgers.edu/riul/research/code/EDISON/index.html. As shown in Fig. 5, the k-means and mean shift algorithms based on the image pixels run quicker than other approaches since the image size is small. However, their segmentation results are not favorable, as shown in Figs. 1, 3, and 4(d) and (e). As the NCuts, KNN_G, and LLE_G methods have no need of iteration, they run a short time. By contrast, the runtimes of L1_G, MSR_G, and MWSG are similar, which is less than LR_G.

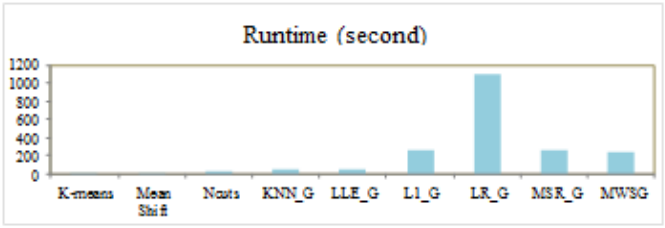


Fig. 5. Runtime (second) of different methods for the 256×256 image.

C. Segmentation of Ku-Band and X-Band SAR Images

To validate the applicability of MWSG-based segmentation algorithm on real SAR images, experiments on two Ku-band and X-band SAR images are implemented. As the ground truth of the real SAR image is generally absent, the evaluation of the segmentation results of different methods is based on visual inspection. Moreover, several existing SAR image segmentation methods are used to compare with the proposed MWSG, which include context-based hierarchical unequal merging for SAR image segmentation (CHUMSIS) [46], higher order neighborhood-based triplet Markov fields model (HN_TMF) [47], and nonlocal fuzzy clustering algorithm with between-cluster separation measure (NS_FCM) [5].

Since the segmentation result obtained by CHUMSIS is the boundary image, we label per region in the segmentation result according to the input image for comparison purposes. The fuzziness index, the parameter α, the size of the neighbor window, and the size of the search window are set to 2, 6, 5×5, and 11×11 for NS_FCM. The parameter setting is the same as that of Section V-B for the other algorithms.

Fig. 6(a) shows a Ku-band SAR image with 3-m spatial resolution of the China Lake Airport, CA, USA. Fig. 6(a) contains three kinds of land cover: airport buildings (bright), vacancy area (gray), and runway (dark). The segmentations of k-means, HN_TMF, NS_FCM, NCuts, KNN_G, LLE_G, and LR_G have many spots in consistent regions, as shown in Fig. 6(b), (e)–(i), and (k), respectively. The segmentation result of the mean shift with $(h_s, h_r, W) = (17, 10, 10)$ shown in Fig. 6(c) has a good visual effect, but it actually contains 103 categories. Although Fig. 6(d) obtained by CHUMSIS is ideal, a small part of the runway in the upper-middle

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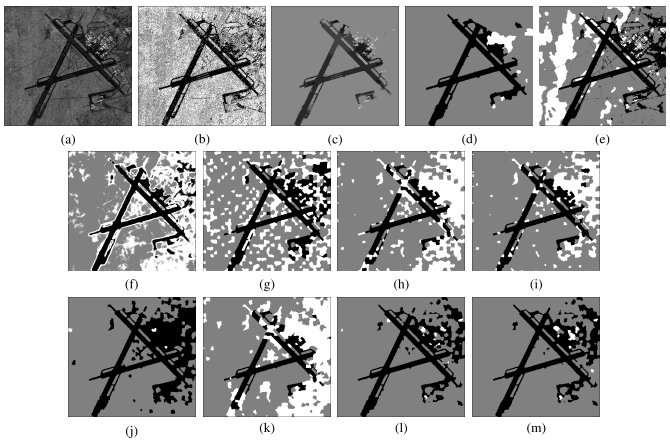


Fig. 6. Segmentation results on the SAR image of China Lake Airport. (a) Ku-band SAR image of China Lake Airport (446 × 475 pixels). (b) k-means. (c) Mean shift. (d) CHUMSIS. (e) HN_TMF. (f) NS_FCM. (g) NCuts. (h) KNN_G. (i) LLE_G. (j) L1_G. (k) LR_G. (l) MSR_G. (m) MWSG.

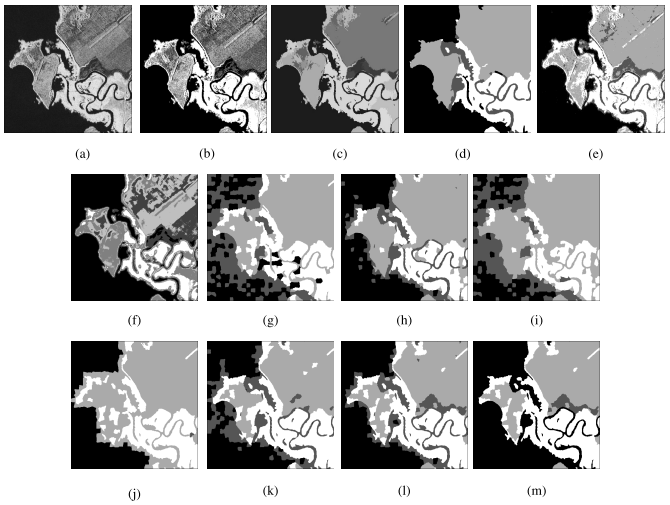


Fig. 7. Segmentation results on the SAR subimage of Russian. (a) X-band SAR subimage of Russian (512 × 512 pixels). (b) k-means. (c) Mean shift. (d) CHUMSIS. (e) HN_TMF. (f) NS_FCM. (g) NCuts. (h) KNN_G. (i) LLE_G. (j) L1_G. (k) LR_G. (l) MSR_G. (m) MWSG.

part of Fig. 6(d) is not separated from the vacancy area. Fig. 6(j) and (l) shows the segmentation results of L1_G and MSR_G, respectively, which has good region consistency. However, some pixels of the vacancy region are misclassified in Fig. 6(j) and (l). In contrast, the segmentation of MWSG shown in Fig. 6(m) not only has regional consistency but also locates accurately the airport buildings area.

Another experiment is executed on a X-band SAR subimage with a 15-m spatial resolution of the south Russian Steppes about 500-km northeast of the Black Sea in 2007, as shown in Fig. 7(a). This SAR image includes four types of ground objects with bare land (bright), crop (light gray), vegetation (dark gray), and water (dark). Fig. 7(b) shows the segmentation result of k-means, in which the regions except the water area have many spots. Similar to Fig. 6(c), though the segmentation of mean shift with $(h_s, h_r, W) = (17, 10, 10)$ has good visual effect, it actually includes 252 categories, as shown in Fig. 7(c). Fig. 7(d) obtained by CHUMSIS has

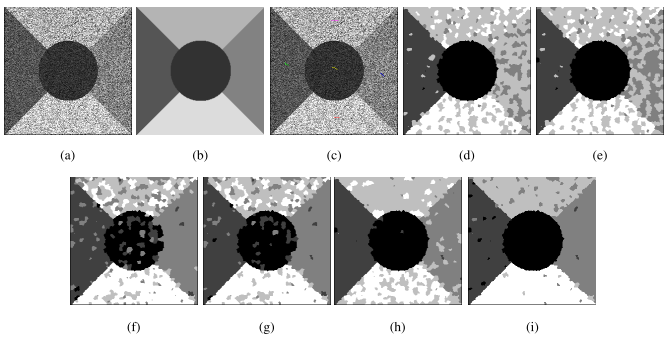


Fig. 8. Semisupervised classification results on five-class artificial image. (a) Artificial image (256 × 256). (b) Ground truth. (c) Labeled artificial image. (d) KNN_G. (e) LLE_G. (f) L1_G. (g) LR_G. (h) MSR_G. (i) MWSG.

favorable regional consistency, whereas the small water and vegetation regions are also confused. The segmentation results of HN_TMF and NS_FCM have poor regional consistency, as shown in Fig. 7(e) and (f). In the segmentations of NCuts, KNN_G, LLE_G, LR_G, and MSR_G shown, respectively, in Fig. 7(g)–(i), (k), and (l), many pixels in the water area are misclassified. Fig. 7(j) cannot distinguish the vegetation from the crop, which is obtained by L1_G. The segmentation obtained by MWSG gives accurate boundaries among the bare land, crop, vegetation, and water, and has good regional consistency, as shown in Fig. 7(m).

D. Semisupervised Classification of Simulated and X-Band SAR Image

In this section, MWSG-based semisupervised classification experiments are carried out on simulated and X-band real SAR images, which further indicates the effectiveness of the proposed MWSG. The parameter α in Algorithm 3 is set to 0.95, and other parameters are the same as those of Section V-C.

As can be seen from Fig. 4(l), two regions from different classes are confused. Fig. 4(a) is used as the simulated SAR image in the semisupervised classification experiment, as we hope that MWSG-based experimental result on Fig. 4(a) is favorable. Fig. 8 shows the semisupervised classification results of graph-based different methods on a five-class simulated SAR image. The simulated SAR image and corresponding ground truth are shown in Fig. 8(a) and (b), respectively. Fig. 8(c) shows the labeled pixels by using different colors for different classes. The classification results of KNN_G and LLE_G confuse two regions from different classes, which are shown in Fig. 8(d) and (e), respectively. In Fig. 8(f)–(h) obtained, respectively, by L1_G, LR_G, and MSR_G, different five regions are separated, but their regional consistency is poor. Fig. 8(i) shows the classification results of MWSG, which has better regional consistency than other methods.

Furthermore, a group of semisupervised classification experiment on a real SAR image shown in Fig. 9(a) is implemented. Fig. 9(a) shows a three-look X-band SAR subimage of Switzerland obtained by Space Radar Laboratory Missions in 1994, which consists of three kinds of land cover: urban (bright), mountain (gray), and water (dark). The labeled pixels are shown in Fig. 9(b) by using different colors for

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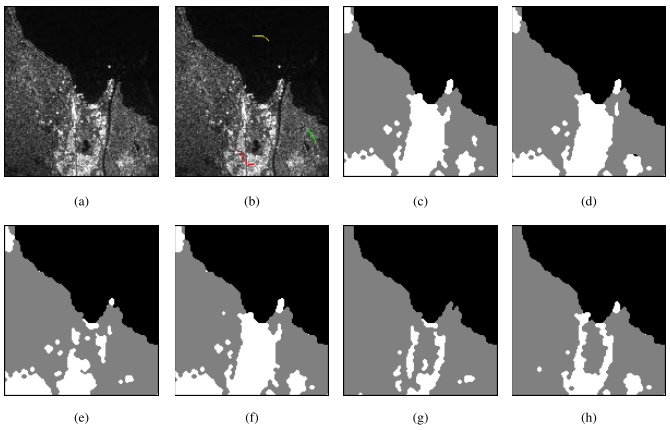


Fig. 9. Semisupervised classification results on the SAR subimage of Switzerland. (a) X-band SAR subimage of Switzerland (155×140). (b) Labeled SAR image. (c) KNN_G. (d) LLE_G. (e) L1_G. (f) LR_G. (g) MSR_G. (h) MWSG.

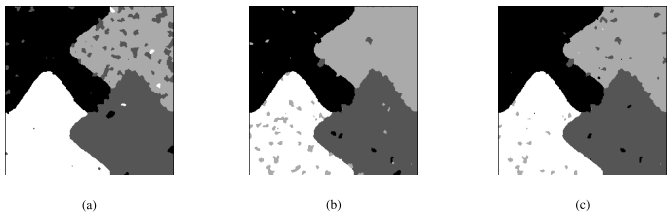


Fig. 10. Segmentation results of different methods. (a) MSR_G (accuracy: 0.9461). (b) Weighted MSR_G (accuracy: 0.9611). (c) MWSG (accuracy: 0.977).

different classes. In the left region of Fig. 9(c)–(f) obtained, respectively, by KNN_G, LLE_G, L1_G, and LR_G, many pixels are misclassified into the urban area. Fig. 9(g) shows the classification of MSR_G, in which a portion of the urban area is only separated from the mountain. The classification result of MWSG is favorable, which locates accurately the urban area, as shown in Fig. 9(h).

E. Discussion

In this article, the weighted way and new multi-feature fusion mechanism are used to improve the category distinguishing the performance of the sparse graph. The effectiveness of two operations is validated by a group of experiment on the four-class simulated SAR image Fig. 3(a). Fig. 10(a) shows the segmentation result of MSR_G which does not contain the weighted way and new multi-feature fusion mechanism. The accuracy of the segmentation result of weighted MSR_G shown in Fig. 10(b) is higher than that of MSR_G. After adding a new fusion mechanism, some singular data are removed, so the segmentation result is improved, as shown in Fig. 10(c). It is can be seen from Fig. 10 that the weighted way and new fusion mechanism are helpful to improve the performance of the sparse graph.

Furthermore, the proposed MWSG includes an important parameter λ. Taking the four-class simulated SAR image Fig. 3(a) as an example, Fig. 11(a) shows the influence of different parameters in MWSG for the segmentation results. As you can see from Fig. 11(a), the segmentation accuracy first increases and then decreases with the increasing of λ, which

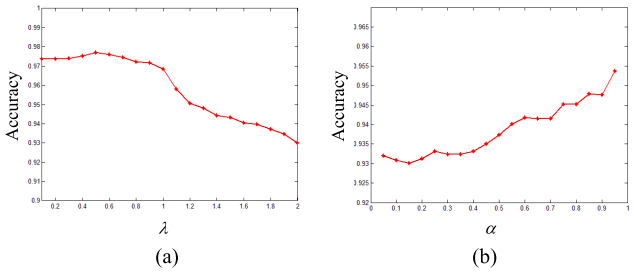


Fig. 11. Influence of different parameters for the results. (a) Parameter λ. (b) Parameter α ($\lambda = 0.5$).

reflects that the sparse term and the fidelity term in weighted joint SR model (4) play the important roles. When the effect of one term in (4) is much greater than another term, the final segmentation accuracy is not best. In addition, MWSG-based semisupervised classification result is affected by the parameter α in Algorithm 3. Take the five-class simulated SAR image Fig. 4(a) as an example. The influence of different values of parameter α on the classification results is shown in Fig. 11(b). As can be seen from Fig. 11(b), the classification accuracy decreases with the increasing of α. It indicates that the more the proportion of MWSG, the higher the final classification accuracy is. This is consistent with the fundamental of semisupervised classification method-based graph, which only needs a small number of label information and employs the connection weights of graph to get good classification result. Meanwhile, it is can be seen from Fig. 11(b) that the proposed MWSG has favorable distinguishing performance.

VI. Conclusion

In this article, a new MWSG was presented for the SAR image segmentation and classification. At first, to fully describe characteristics of SAR image, multiple kinds of SAR image features are extracted. Also, the Gaussian kernel distance between every two samples in each feature space is calculated as the weight. And then, a new weighted joint SR model was given and used to get the SRs of samples in multiple feature spaces. Considering the negative influence of the singular data, some singular data are removed in the progress of fusing multiple weighted SRs, which is helpful to improve the performance of MWSG. By this fusion operation, multiple weighted SRs in multiple feature spaces are integrated to construct the MWSG. Finally, MWSG-based SAR image segmentation and semisupervised classification experiments on the simulated and real SAR images show that the performance of MWSG is improved due to the usage of the weighted way and new fusion mechanism.

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